

ECEN-601: Homework 7

Due on November 26, 2025 at 11:59 pm

Instructor: Dr. Jean-François Chamberland

Brody Jordan
431005357

Problem 1

Let V be a finite-dimensional inner product space and T a linear operator on V . Show that the range of the adjoint T^* is the orthogonal complement of the null space of T .

Solution

Problem 2

If P is an orthogonal projection matrix, show that $I - P$ is a orthogonal projection matrix. Determine the range and nullspace of $I - P$.

Solution

Problem 3

Show that, if A has linearly independent columns, then

$$B = I - A(A^H A)^{-1} A^H$$

is positive semidefinite, and hence that the minimum error $\mathbf{e}_{\min} = \mathbf{x} - P_A \mathbf{x} = B\mathbf{x}$ has an equal or smaller norm than the original vector $\mathbf{x}$. Hint: Consider $\|\mathbf{e}_{\min}\|^2 \geq 0$.

Solution

Problem 4

Formulate and solve the abstract least-squares regression problem

$$\min_{a_0, a_1, a_2} \sum_{i=1}^n |y_i - a_0 - a_1 x_i - a_2 x_i^2|^2$$

in its linear form based on $\vec{e} = A\vec{c} - \vec{y}$. Use this to fit a parabola to the (x_i, y_i) data points $(-2, 2)(-1, -10)(0, 0)(1, 2)(2, 1)$. Make a plot showing the data on top of the fitted curve.

Solution

Problem 5

Define an inner product between matrices $X, Y \in \mathbb{C}^{l \times n}$ as

$$\langle X, Y \rangle = \text{tr}(XY^H),$$

where $\text{tr}(\cdot)$ is the sum of the diagonal elements (see section C.3). We want to approximate the matrix Y by the scalar linear combination of matrices $X_1, X_2, \dots, X_m$, as

$$Y = c_1 X_1 + c_2 X_2 + \dots + c_m X_m + E.$$

Using the orthogonality principle, determine a set of normal equations that can be used to find $c_1, c_2, \dots, c_m$ that minimize the induced norm of E .

Solution

Problem 6

Using the dual approximation theorem, solve the finite dimensional problem

$$\begin{aligned} \text{minimize: } & \mathbf{x}^H Q \mathbf{x} \\ \text{subject to: } & A \mathbf{x} = \mathbf{b}, \end{aligned}$$

where $\mathbf{x} \in \mathbb{C}^n$, Q is a positive-definite symmetric matrix, and A is an $m \times n$ matrix with $m < n$.

Solution